

HYPOTHESIS TESTING FOR DEPENDENT VARIABLES (HTDV)

José Mauricio Gómez Julián

mauriciogomezjulian@protonmail.ch

ORCID: [0009-0000-2412-3150](https://orcid.org/0009-0000-2412-3150)

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1. The Problem It Solves

Picture two everyday situations that look different but are really the same:

(a) You want to know whether two branches of a supermarket sell eggs at the same average price. You have 200 visits to branch A and 60 to branch B (unequal sample sizes). Prices don't move at random from visit to visit: if the price went up today, it's likely still high tomorrow (observations are *temporally correlated*¹). And occasionally a big shock arrives (a sudden inflation spike, a stock-out) producing extreme values (*heavy tails*²).

(b) You want to know whether the 10-year U.S. Treasury yield and the 10-year Canadian yield are statistically equal in the long run. You have 35 years of monthly data (435 observations per country). The series are extremely persistent³: this month's yield is tightly tied to last month's. And they aren't normal⁴ either: heavy tails from interspersed financial crises.

In both cases the standard method taught in introductory statistics courses — the classic *Student's t-test*⁵ or its cousin Welch's t-test⁶ — **gives a numerical answer, but the answer is not trustworthy**. Why? Because those methods assume each

¹ Saying two observations are **temporally correlated** means that knowing today's value gives you information about tomorrow's value. If the data were independent, knowing today's price would tell you nothing about tomorrow's; with positive temporal correlation, a high price today makes a high price tomorrow more likely.

² A distribution has **heavy tails** when values far from the mean appear more frequently than a normal distribution would predict. In practice: outliers and extreme values are not exceptional rarities, they occur regularly.

³ **Persistence** measures how strongly a series "remembers" its recent past. A highly persistent series changes slowly: if it's high today, it will probably still be high tomorrow, the day after, and next week. Formally it is quantified through the autocorrelation $\rho_k = \text{Corr}(X_t, X_{t+k})$.

⁴ A variable is **normal** (Gaussian) if it follows the familiar symmetric bell curve in which 95% of the data lies within 2 standard deviations of the mean. Many classical methods are built assuming normality, and they fail silently when it does not hold.

⁵ **Student's t-test** is the classical method for deciding whether the mean of a sample differs from a reference value (or whether the means of two samples differ from one another), under the assumptions that the data are independent, identically distributed, and approximately normal.

⁶ **Welch's t-test** is a variant of the t-test that does not require the two samples to share the same variance. It still requires independence and approximate normality.

observation is independent and identically distributed (*i.i.d.*⁷). When there is dependence, unequal sample sizes, or heavy tails, the test-statistic formula keeps cranking out numbers, but the p-value⁸ and confidence interval⁹ it produces no longer mean what they claim to mean. You say “I reject with 5% type I error¹⁰” but in reality you are rejecting with 30% or 60% error. The method’s calibration¹¹ is broken.

HTDV was built precisely for this scenario: testing hypotheses about means (one sample against a value, or between two samples) when the classical assumptions are broken, and delivering conclusions whose calibration can be *empirically demonstrated* – not assumed – under dependence, imbalance, and heavy tails.

2. Why The Classical Solution Fails?

The classical t-test assumes three things:

- **Independence:** observations don’t influence each other.
- **Identical distribution:** they all come from the same unknown distribution.
- **Finite variance:** the second moment¹² exists.

If all three hold, the classical **Law of Large Numbers (LLN)**¹³ guarantees that the sample mean converges to the population mean, and the **Central Limit Theorem**

⁷ **i.i.d.** stands for “independent and identically distributed.” Independence means that the value of one observation doesn’t influence the others; identically distributed means they all come from the same random mechanism.

⁸ The **p-value** is the probability of observing a result at least as extreme as the one observed, assuming the null hypothesis is true. If small (typically < 0.05), the null is rejected.

⁹ A 95% **confidence interval** is a range built so that, if we repeated the experiment many times, 95% of those ranges would contain the true parameter value. It is *not* the probability that the parameter lies in that particular range.

¹⁰ A **type I error** is rejecting the null hypothesis when it is in fact true (a “false positive”). The nominal level $\alpha = 0.05$ means you are willing to commit this error 5% of the time.

¹¹ A method is well **calibrated** when the rate at which it commits type I errors in practice matches the rate it claims to commit. A poorly calibrated method says “I’m wrong 5% of the time” but actually is wrong 30% of the time.

¹² The **second moment** of a variable is $E[X^2]$. The variance is the second centered moment: $\text{Var}(X) = E[(X - \mu)^2]$. Finite variance means extreme values are not so frequent as to make the average of the squares blow up.

¹³ The **Law of Large Numbers** states that the average of many observations approaches the theoretical mean of the distribution as the number of observations grows. It is the mathematical justification for why flipping a coin many times yields roughly 50% heads.

(CLT)¹⁴ guarantees that the error of that mean is approximately normal with variance σ^2/n . These two theorems are the raw material of everything taught in introductory statistics courses.

When independence breaks, the variance of the sample mean is no longer σ^2/n but something of order σ_∞^2/n , where σ_∞^2 is the **long-run variance**¹⁵ – which sums all autocovariances¹⁶, not just the pointwise variance. If you keep using σ^2/n to build your interval, the interval comes out artificially narrow. Falsely precise. This is exactly the problem documented at industrial scale by the simulation in `htdv_simstudy()`: under strong dependence, methods calibrated with the pointwise variance err in the direction of “I know more than I actually know.”

3. The Conceptual Core: The LLN And Its Non-Classical Variants

The classical LLN says something simple: if you flip a coin many times, the proportion of heads approaches 1/2. More formally, for an i.i.d. sequence $X_1, X_2, \dots$ with finite expectation μ , the sample mean

$$\bar{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n}$$

converges to μ as $n \rightarrow \infty$.

There are two senses in which it can converge:

- **Weak version**¹⁷: the mean approaches μ with high probability when n is large, but excursions away from μ are still possible. Convergence in probability: $\bar{X}_n \xrightarrow{P} \mu$.

¹⁴ The **Central Limit Theorem** states that, under reasonable conditions, the distribution of the sample mean – appropriately rescaled – looks more and more like a normal distribution as the sample size grows, regardless of the original distribution of the data.

¹⁵ The **long-run variance** σ_∞^2 is the correct variance of the sample mean when observations are temporally correlated. It sums not only the variability of each observation but also all covariances between observations close in time. If the data were i.i.d., σ_∞^2 coincides with σ^2 .

¹⁶ The **autocovariance** at lag k is $\gamma(k) = \text{Cov}(X_t, X_{t+k})$: it measures how much the series co-varies with itself k steps apart. The pointwise variance is the case $k = 0$.

¹⁷ **Convergence in probability** ($\xrightarrow{P}$) means that for any error margin $\varepsilon > 0$, the probability that the sample mean strays from the true value by more than ε tends to zero as n grows. It allows occasional deviations, provided they become rarer and rarer.

- **Strong version**¹⁸: the mean actually settles near μ from some n onward, with no further excursions. Almost-sure convergence: $\bar{X}_n \xrightarrow{a.s.} \mu$.

For i.i.d. data, the difference between the two versions is technical. But **when there is dependence, the difference becomes substantive**: the tools that prove the strong version are not the same as the ones that prove the weak version, and the assumptions about dependence each one requires are different.

The classical LLN is a single theorem that covers the easy case (i.i.d.). The non-classical LLNs are a *family* of theorems, each targeting a specific mechanism through which dependence might break convergence. Each theorem is “local” in the sense that it captures a precise class of dependence (not all dependence is alike). Together they cover the difficult cases.

4. The Three Versions The Package Studies

HTDV picks three non-classical LLNs, each useful for a different type of dependence that arises in real data.

4.1. TAC – Triangular Array Convergence (Strong Version)

Triangular arrays¹⁹ are doubly indexed sequences: row k has k entries. They are the natural model when the data are **hierarchical aggregations**: an input-output matrix where each sector subdivides into subsectors, each subsector into activities, each activity into products. As you drill deeper into the hierarchy, the “row” grows longer.

The strong LLN for triangular arrays says: under α -mixing²⁰ with polynomial decay $\gamma > 1$ and finite moments of the appropriate order, the within-row mean converges almost surely to the population mean as the row grows long.

¹⁸ **Almost-sure convergence** ($\xrightarrow{a.s.}$) is stricter: it says that, except on a set of events of probability zero, the sample mean eventually sticks to the true value from some n onward. It’s not just unlikely to stray: it stops straying altogether.

¹⁹ A **triangular array** is a collection of random variables arranged in rows: $\{X_{n,1}, X_{n,2}, \dots, X_{n,n}\}_{n \geq 1}$. Row n has n elements, and the joint distribution of the row may change with n . It is the natural tool for studying averages whose observations change in nature as the sample size grows (for example, increasingly fine-grained observations).

²⁰ A sequence is **α -mixing** if the dependence between the distant past and the distant future weakens as they separate in time. Formally, the coefficient $\alpha(k)$ measures how “far” events k steps apart can be from being independent. Polynomial decay $\gamma > 1$ means $\alpha(k)$ falls at least as fast as $k^{-\gamma}$.

This is useful when you have hierarchical data and want to know whether the aggregate converges to the true parameter as you increase the resolution.

4.2. WSC – Weighted Sums with Correlation (Strong Version)

The classical LLN computes the sample mean by giving each observation the same weight $1/n$. WSC allows **different weights** $\omega_1, \omega_2, \dots$ that reflect the **correlation structure** between observations:

$$S_n = \sum_{i=1}^n \omega_i X_i, \quad \omega_i = \frac{1}{n} \cdot \frac{1}{1 + \sum_j \rho_{ij}}.$$

The weight on observation i shrinks when it is highly correlated with its neighbors (because it then contributes less new information). This is exactly the logic of expected value as a probability-weighted sum, generalized to the case where “probability” is replaced by “information conditional on correlations.”

This is useful when dependencies between observations are explicitly correlational (one economic sector depends on another via suppliers; two countries’ yields correlate through financial integration; two analysts measuring the same sample share protocol noise).

WSC is the most direct way of extending the logic of expected value to non-i.i.d. data, which makes it philosophically appealing: you are not changing the nature of the estimator²¹, you are formally recognizing that some observations count for less because they are correlated with others.

4.3. MPC – Mixingale Processes Convergence (Weak Version)

A **mixingale**²² is a temporal process where the effect of the past on the present decays at a controllable rate. Formally, if $\mathcal{F}_{t-k}$ is the information up to k steps back, then

$$\| E[X_t | \mathcal{F}_{t-k}] - E[X_t] \|_2 \leq \psi_k,$$

²¹ An **estimator** is a recipe for computing a number (such as the sample mean) from data, with the goal of approximating an unknown parameter of the population (such as the population mean). An estimator is good if, as the sample grows, it homes in on the true parameter (consistency) and if, in small samples, it does not err systematically (unbiasedness).

²² A **mixingale** is a technical object that generalizes both martingales and α -mixing processes: it captures the idea that the best prediction of the present value, given far-past information, should look more and more like the unconditional mean. The rate ψ_k controls how quickly the series’ memory “fades.”

where $\psi_k \rightarrow 0$ at a rate you can bound.

The LLN for mixingales (weak version) says: under conditions on the weights ψ_k , the weighted sum $\sum \psi_t(X_t - \mu)$ has bounded variance, so the normalized sum converges in probability to zero.

This is useful when observations are temporally dependent but the dependence is forgotten over time: forecast errors that don't accumulate indefinitely, residuals²³ from time-series models, prediction intervals that gradually lose power.

4.4. Why Three Flavors Of The LLN and Not One?

Each regime captures a different mechanism by which real data may violate i.i.d.:

- **TAC**: *hierarchical* dependence. Information accumulates as you subdivide.
- **WSC**: *correlational* dependence. Information shares covariance structure.
- **MPC**: *temporal dependence with decaying memory*. Information is forgotten.

A single LLN that covered all three would be so general that its finite-sample constants²⁴ would be unusable. Three specific LLNs are three sharp tools, each with an explicit constant useful for real decisions. The immediate question, of course, is: aren't they then three parallel frameworks that give different answers to the same problem?

That is where the result that ties the apparatus together comes in.

5. The Unifying Theorem (Strict Metric Equivalence)

The theorem says something strong: under α -mixing with $\gamma > 1$ and finite moments, the **distances**²⁵ D_T^{TAC} , D_T^{WSC} , and D_T^{MPC} that each regime induces over the space of hypothesis tests are **strictly equivalent**: there exist explicit constants

²³ **Residuals** are the difference between what is observed and what the model predicts: $e_t = X_t - \hat{X}_t$. If the model captured the structure of the data well, the residuals should behave like noise (with no systematic pattern).

²⁴ **Finite-sample constants** are the explicit numbers that show up in the error bounds when n is not infinity but, say, $n = 40$ or $n = 100$. Asymptotic theorems give guarantees "for n large enough," which is useless if you don't know how large. Finite-sample constants tell you exactly how much error you can have at each concrete n .

²⁵ A **distance** (or metric) between two objects — here, between two hypothesis tests — is a function that measures how "far apart" they are, satisfying properties such as symmetry and the triangle inequality. Two distances are **equivalent** when they yield the same notion of closeness up to a bounded multiplicative factor.

$K_{TAC}, K_{WSC}, K_{MPC}, c_L, c_U$ — which you can compute for any γ, q, n via the function `htdv_equivalence_constants()` — such that

$$c_L \cdot D_T^{(B)} \leq D_T^{(A)} \leq c_U \cdot D_T^{(B)}$$

for any pair of regimes $(A, B) \in \{TAC, WSC, MPC\}$. That is, any result obtained under one regime translates to the others with numerically bounded *slack*²⁶.

In other words, if I apply TAC and you apply WSC to the same problem, we are not comparing apples to oranges; we are comparing things that differ by a computable constant, and that constant converges to 1 as the sample size grows. Asymptotically the three metrics are the same; in finite samples, they are the same with explicit error.

This is what makes the entire framework **mathematically well-defined**, not a collage of tools. Without metric equivalence, mixing HAR²⁷ with a Bayesian TAC fit would be incoherent. With metric equivalence, the two methods are measuring the same thing and can be compared.

6. The Three Inferential Layers (The Practical Machinery)

Now comes the executable part. Once the theoretical apparatus is granted, **HTDV** delivers three independent ways to compute a confidence/credibility interval for the parameter of interest.

6.1. Layer 1 — Bayesian (HMC via Stan)

It builds a parametric likelihood²⁸ for the data (Whittle²⁹ in the frequency domain, or composite in the time domain), places *weakly informative* priors³⁰ on the nuisance

²⁶ **Slack** is the allowed error or gap between two quantities. That the slack is explicit means it isn't a vague "give or take," it's a concrete number you can compute and report.

²⁷ **HAR** ("Heteroskedasticity and Autocorrelation Robust") refers to a family of methods for estimating the variance of an average or coefficient when the errors are heteroskedastic (changing variance) and autocorrelated (time-dependent). At its core is a long-run variance estimator built from kernels (Bartlett, Parzen, etc.) and a bandwidth controlling how many autocovariances are included.

²⁸ The **likelihood** is a function that, given fixed data, returns how compatible different parameter values are with that data. If $f(x | \theta)$ is the data density given θ , the likelihood is $L(\theta) = f(x_{\text{obs}} | \theta)$ viewed as a function of θ . **Parametric** means you assume a specific family of distributions (for example, normals with some mean and variance).

parameters³¹ (σ_∞, γ, b), and samples the posterior³² with Hamiltonian Monte Carlo³³. From the posterior it computes:

- **ROPE** (Region of Practical Equivalence)³⁴ for decision-making.
- **Bayes factor**³⁵ via bridge sampling for model comparison.
- **WAIC**³⁶ and **leave-future-out**³⁷ for predictive evaluation.
- **Stacking**³⁸ for averaging predictives.

This layer scales best with model complexity and is the only one that delivers a full *posterior* (not just a point and an interval).

²⁹ The **Whittle likelihood** is an approximation to the likelihood of a stationary time series operating in the frequency domain (via the Fourier transform) instead of the time domain. It is much cheaper computationally for long series and captures the autocovariance structure well.

³⁰ In Bayesian statistics, a **prior** distribution represents knowledge (or uncertainty) about the parameters before seeing the data. *Weakly informative* means it does not commit to any specific value but rules out absurd ones.

³¹ **Nuisance parameters** are those that are not the direct object of interest, but must be estimated anyway to make inference about the ones that are. For example, if you care about the mean, the variance is a nuisance parameter.

³² The **posterior** distribution is the update of the prior after incorporating the data, via Bayes' theorem: $p(\theta | x) \propto L(\theta) \cdot p(\theta)$. It summarizes everything you know about the parameters once the data have been seen.

³³ **Hamiltonian Monte Carlo** is an algorithm for sampling from complicated distributions (such as a high-dimensional Bayesian posterior) using an analogy with classical mechanics: it simulates the motion of a "particle" over the log-density landscape. It is far more efficient than older Monte Carlo methods on high-dimensional problems.

³⁴ The **Region of Practical Equivalence** (ROPE) is an interval around the null hypothesis value that is considered "practically indistinguishable" from the null value. The Bayesian decision is taken by comparing the posterior against this region: if the posterior falls mostly outside, reject; if it falls inside, accept.

³⁵ The **Bayes factor** between two models is the ratio of their marginal likelihoods, and measures how much more probable one model makes the data than the other. It is the Bayesian counterpart of the likelihood ratio.

³⁶ **WAIC** (Widely Applicable Information Criterion) is a measure of how well the Bayesian model predicts new data, computed from the posterior. It penalizes model complexity to avoid overfitting.

³⁷ **Leave-future-out** is the time-respecting version of cross-validation: it fits the model with data up to time t , predicts $t + 1$, and evaluates. It does not peek into the future when fitting.

³⁸ **Stacking** is a technique for combining predictions from several models by assigning them optimal weights according to their predictive performance, instead of choosing one or naively averaging them.

6.2. Layer 2 – Frequentist HAR (Long-Run Variance)

No likelihood needed. It estimates the long-run variance σ_{∞}^2 with a HAC kernel³⁹ (Bartlett, Parzen, Quadratic-Spectral) and an optimal Andrews bandwidth, then constructs a Wald statistic⁴⁰

$$W = \frac{n(\bar{X} - \theta_0)^2}{\sigma_{\infty}^2},$$

with critical values calibrated under the *fixed-b* asymptotics of Kiefer-Vogelsang⁴¹ (which correct the finite-sample distortion of standard HAR). It is fast, requires no model specification, and delivers a classical p-value.

6.3. Layer 3 – Stationary Block Bootstrap

No likelihood needed either. It takes the observed series and **resamples it in blocks**⁴² – not observation by observation, but blocks of random geometric length – in order to preserve the dependence structure. The optimal block length is automatically chosen by the Patton-Politis-White rule. From the $R = 1999$ resamples a distribution-free⁴³ percentile interval is produced.

6.4. How The Three Are Used Together?

The philosophy is not “pick one.” It is **run all three and read the disagreement as information**:

³⁹ **HAC** (“Heteroskedasticity and Autocorrelation Consistent”) is a family of estimators of the long-run variance that builds a weighted average of the sample autocovariances. The weight function – the “kernel” – and the bandwidth (how many lags to include) are the two critical choices.

⁴⁰ A **Wald statistic** has the form $(\hat{\theta} - \theta_0)^2 / \widehat{\text{Var}}(\hat{\theta})$: it measures the distance between the estimator and the null value, normalized by the estimator’s uncertainty. Under the null and regularity conditions, it follows approximately a chi-squared distribution.

⁴¹ The **fixed-b** asymptotics treats the HAC kernel bandwidth as a fixed fraction of the sample size (rather than letting it tend to infinity slower than n). The result is that the test statistic no longer converges to a chi-squared, but to a non-standard yet known distribution, and the critical values tabulated under this asymptotics have better small-sample performance.

⁴² The **bootstrap** is a technique that approximates the distribution of an estimator by resampling the data with replacement and recomputing the estimator on each resample. The **block bootstrap** modifies it for time series: instead of resampling individual observations (which would destroy dependence), it resamples contiguous blocks of the series, thus preserving the local dependence structure.

⁴³ A method is **distribution-free** when it does not require assuming the data come from any specific family of distributions (normal, exponential, etc.). Its guarantees hold under much weaker assumptions.

- **If all three agree:** the conclusion is robust. Make the decision.
- **If Bayes is much wider than HAR/bootstrap:** dependence is probably high and the frequentist asymptotics are failing. Trust Bayes.
- **If HAR/bootstrap is much wider than Bayes:** the Bayesian likelihood is probably mis-specified. Investigate the specification.
- **If the two frequentist layers agree with each other but diverge from Bayes:** a key clue to which model assumption is propping up the result.

This is the *very point* of the package: not to give you a single answer you sign off on blindly, but three independent answers you cross-check to understand how solid the inference is.

7. The Validation Rvidence

That all three layers agree under i.i.d. is trivial; any framework should pass that test. What matters is showing it **works where the classical methods fail**. The package ships with two pieces of evidence.

7.1. The Factorial Simulation (Internal Validation)

`htdv_simstudy()` runs a full factorial design⁴⁴ of 1024 cells:

- 4 sample sizes $\times$ 4 persistence levels $\times$ 4 tail distributions $\times$ 4 imbalance ratios $\times$ 4 shift magnitudes = $4^5 = 1024$ cells.
- 500 Monte Carlo replicates⁴⁵ per cell $\times$ 3 layers = 1.5 million inferential runs.
- 31 hours of wall-clock time on 16 cores.

The datasets `htdv_sim_summary` and `htdv_empirical_benchmarks` shipped with the package contain the digested results. **Central finding:** at nominal size $\alpha = 0.05$, the Bayesian layer attains an empirical mean size of 0.056 across the entire grid; HAR attains 0.18 ($3.6 \times$ the nominal); bootstrap 0.18. Under persistence $\phi = 0.85$ and $n = 40$, HAR rejects 60% of the time when it should reject 5%.

This is the numerical demonstration that the only universally well-calibrated layer is the Bayesian one.

⁴⁴ A full **factorial design** is an experiment where every possible combination of several factors at their different levels is tested. Here: 5 factors $\times$ 4 levels each = $4^5 = 1024$ distinct combinations, each replicated 500 times.

⁴⁵ A **Monte Carlo replicate** is a simulated repetition of the experiment under the same scenario, generating new data from the same random mechanism each time. The average of results across replicates approximates the expected behavior of the method.

7.2. The Empirical Benchmarks (External Validation)

Three public datasets, three published references, three **agreement** verdicts:

- **FRED-MD CPI inflation post-1984** vs. Stock & Watson 2007 (annualized mean inflation $\approx 2.7\%$).
- **Shiller log-CAPE 1881–present** vs. Campbell & Shiller 1998 (historical mean ≈ 2.85).
- **U.S.–Canada 10-year bond differential 1990–present** vs. Welch i.i.d. baseline.

All three references fall inside the 95% interval that HTDV produces. And the substantive finding: the *width* of the intervals grows monotonically with the persistence of the series. At moderate persistence, the Bayesian intervals coincide with the HAR ones. At persistence near the unit root⁴⁶, the Bayesian intervals are **15× wider** than the HAR ones. That difference isn't Bayes being conservative: it is Bayes being honest about real uncertainty under near-non-stationarity⁴⁷.

8. The Package As Coherent Architecture

What HTDV delivers is not a new statistical test. Each individual component exists in the literature: Whittle 1953, Newey–West 1987, Andrews 1991, Kiefer–Vogelsang 2005, Politis–Romano 1994, Patton–Politis–White 2009, Gibbs–Candès 2021, Vehtari et al. 2017, Kruschke 2018. What the package delivers is:

- **A theorem unifying three non-classical LLNs in a single metric apparatus**, with explicit finite-sample constants computable via `htdv_equivalence_constants()`. This is original contribution and is what makes the rest of the apparatus coherent.
- **A three-layer inferential architecture** (Bayesian, HAR, bootstrap) built on top of that apparatus, where disagreement between layers is information, not error.
- **Its own internal and external validation evidence** packaged as datasets accessible from the R console: `data(htdv_sim_summary)`;

⁴⁶ A series has a **unit root** when it is non-stationary in a very specific way: shocks never die out but are permanently absorbed into the level of the series. The textbook case is the random walk $X_t = X_{t-1} + \varepsilon_t$. Near the unit root, classical inference becomes extremely misleading.

⁴⁷ A series is **stationary** if its statistical properties (mean, variance, autocovariances) do not change over time. *Near-non-stationarity* describes series on the verge of losing stationarity: technically stationary, but so close to the unit root that they behave empirically as non-stationary in finite samples.

`data(htdv_empirical_benchmarks)`. This is what makes the package *auditable*: you don't ask the user to trust; you show them the 3069 rows of evidence.

- **A complete reproducibility infrastructure** in `data-raw/`: the executable scripts that generated the simulation, the empirical benchmarks, and the post-hoc correction of the sign bug. Anyone with WSL and 16 cores can regenerate the evidence from scratch.
- **Everything under CRAN⁴⁸ compliance** (with the operational caveat of the current `.Rbuildignore` issue in your environment), with vignettes navigable via `vignette("HTDV-validation")` and canonical citation via `citation("HTDV")`.

9. The One-Line Explanation

It enables hypothesis testing about means when the data are dependent, imbalanced, or heavy-tailed – scenarios where the t-test lies about its own precision – . It does so by combining three independent ways of computing confidence intervals (one Bayesian, one frequentist, one distribution-free), tied together by a theorem proving that all three measure the same thing under explicit conditions. And it ships with the empirical evidence that the method works where the classical ones fail.

For the mathematical audience, the `htdv_equivalence_constants()` function should be used to determine the finite sample constants. For the applied statistical audience, calibration is carried out using the `data(htdv_sim_summary)` function. For the student audience, see section 1 of this document (the example of the eggs in branch A versus branch B).

We consider the strength of the package to be that all three conversations are supported by the same underlying evidence.

⁴⁸ **CRAN** (Comprehensive R Archive Network) is the official repository for R packages. For a package to be accepted, it must pass a strict series of automated checks regarding structure, documentation, executable examples, absence of errors, etc. Subsequently, it must pass a new and final layer of reviews performed by a specialized human reviewer.